SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

SRM Institute of Science and Technology · Academic Year 2026

INSTITUTIONAL MASTER EDITION

Official Production Portal: Hosted on Vercel Cloud `https://electifysrm.vercel.app`

Institutional Operations & Architecture Manual

Institutional Master Manual: Comprehensive architectural specification, security protocols, and operational workflows for SRMIST administrators, academic evaluators, and system developers.

1. Cloud Production Architecture

Component	Technology	Production Endpoint / Role
Frontend Portal	Next.js 16 (React 19, Tailwind v4)	`https://electifysrm.vercel.app`
Backend REST API	NestJS 11 (Express, Swagger UI)	`https://electifysrm.vercel.app/api` (Docs: `/api/docs`)
Database & ORM	PostgreSQL 16, Drizzle ORM	Supabase Transaction Pooler (`aws-1-eu-central-1`)
Multi-Service Routing	Vercel Services & Service Bindings	Internal binding: `NESTJS_API_URL`

2. University Role Credentials Summary

Role	Identifier	Default Password	Authority Scope
System Admin	`srmelectify@gmail.com`	`Electify@2026`	Superuser, Maintenance Mode, Audit Logs.
Course Coordinator	`sabenabanu@srmist.edu.in`	`Electify@2026`	Electives, Capacities, Windows, Surveillance, Replay.
Class Tutor	`angelb1@srmist.edu.in`	`Electify@2026`	Class Rosters, PIN Reset Authorization, Attendance PDF.
Student	SRM Register Number	Student Configured	Subject Explorer, Seat Claim, Receipt Download.

3. Technical Concurrency & Security Safeguards

- **PostgreSQL Row-Locking Concurrency:** Seat reservations use `SELECT ... FOR UPDATE` within ACID transactions. Zero waitlists and zero overbooking.
- **Single-Session Sentinel:** Prevents multi-device sniping by invalidating prior JWT signatures immediately upon new sign-in.
- **NTP Clock Synchronization:** Client countdown timers are synchronized to server NTP time to eliminate client-side device tampering.
- **Tamper-Evident Receipts:** Formatted as `ELC-2026-EVEN-XXXXX` with cryptographic verification signatures.

4. Automated SQA Verification (Zero-Defect Status)

> **Automated Quality Assurance: 20 Tests Passed / 0 Failed (100% Pass Rate)**
> Execute anytime with `npm run test:sqa` to verify Supabase connectivity, 27 schema tables, role guards, proxy routing, and REST endpoints.